

INTELLIGENCE FUSION & INVESTIGATION PLATFORM

TRACE-X

Comprehensive System Architecture, Module Breakdown & Complete Webpage Operational Guide



Multi-Source Intelligence Fusion



Entity Resolution & Graph Analysis



Global Threat Telemetry



Dual-Engine AI Copilot

FULL TECHNICAL DOSSIER

Project: TRACE-X Intelligence Fusion Engine

Scope: Complete Webpage-by-Webpage Explanation & Backend Design

PLATFORM SPECIFICATIONS

Architecture: Node.js + Express + React 18 + PostgreSQL

Release: Enterprise Edition v2.4 (Verified)

1. Executive Summary & Project Purpose

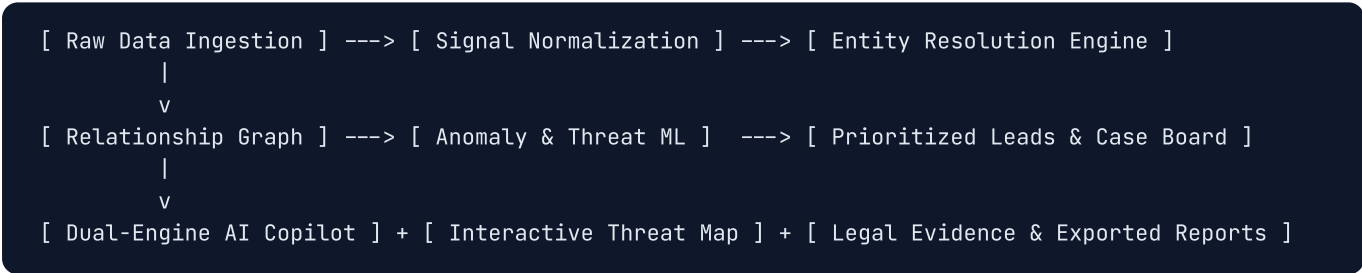
TRACE-X is a next-generation *Intelligence-Fusion and Cyber-Investigation Suite* designed to convert fragmented, high-volume raw signals (network logs, financial wire transfers, dark-web intercepts, domain registrations, crypto ledgers, and communication records) into verified, prioritized investigative leads.

💡 Hindi / Hinglish Summary (TRACE-X Kya Hai Aur Kyon Banaya Gaya Hai?)

Jab bhi koi large cyber attack, financial scam, ya organised crime ki investigation hoti hai, to sara data alag-alag format aur sources me bikhra hota hai (jaise IP logs, Bitcoin wallet hashes, encrypted aliases, phone numbers). Investigator ke liye in lakho dots ko manually connect karna lagbhag asambhav hota hai.

TRACE-X ka kaam hai: In sabhi disconnected signals ko ek central intelligence engine me fuse karna, entity resolution algorithm se suspects ke hidden aliases ko match karna, force-directed graph se relationships visualize karna, aur AI Copilot ki madad se accurate lead ranking provide karna taaki case jaldi solve ho sake with 100% legal evidence traceability.

Core Investigative Workflow



Entity Resolution

AUTOMATED MATCHING

Alag-alag emails, crypto wallets, hardware MAC IDs aur aliases ko ek single verified suspect entity profile se link karta hai.

Graph Link Analysis

MULTI-HOP LINKS

Hidden networks, intermediary bridge nodes, money mules aur shell companies ke connections ko visual node graph me dikhata hai.

AI Analyst Copilot

NATURAL LANGUAGE

Case summaries generate karta hai, highest risk nodes calculate karta hai, aur real-time telemetry queries answer karta hai.

2. Full-Stack Technology Architecture

Layer / Component	Stack & Libraries	Core Technical Role
Frontend UI & Visualization	React 18, Vite, Vanilla JS/TS, Tailwind CSS, HTML5 Canvas, SVG Graphing, Lucide Icons	High-speed SPA router, cyber-themed dark mode, interactive canvas threat map, force-directed graph renderer, real-time alert ticker.

Layer / Component	Stack & Libraries	Core Technical Role
Backend API Server	Node.js, Express.js REST API, Helmet, CORS, JWT Auth, Bcrypt, Dotenv	Business rules, authentication gatekeeper, scoring algorithms, ingestion pipeline handlers, evidence hashing, query routers.
Data Storage Layer	PostgreSQL Relational DB, Sequelize / pg Schema	Structured relational models: Entities, Relationships, Signals, Investigations, Evidence, Alerts, Users & Audit Logs.
Intelligence & Copilot Engine	Dual-Engine TRACE-X AI Assistant, Statistical ML Heuristics	Contextual case synthesis, automated bridge-node scoring, risk classification, query search & dossier extraction.

3. Complete Webpage-by-Webpage & Module Guide

TRACE-X platform ke total **14 core pages aur components** ka comprehensive breakdown neeche diya gaya hai:

3.1. Authentication & Security Clearance ([#login](#) / [#signup](#))

ACCESS CONTROL

Purpose: System ko unauthorized access se secure rakhna aur investigator clearance level verify karna.

- **UI Components:** Cyber-styled login form, Security Clearance status banner, Demo Quick-Login shortcut button, Registration view.
- **Security Measures:** Passwords bcrypt hashing ke sath store hote hain, aur session JWT (JSON Web Token) bearer token ke through manage hota hai.
- **Data Flow:** User credentials enter karta hai -> Backend [/api/auth/login](#) verify karta hai -> Signed JWT return hota hai -> Local state initialize hoti hai -> Main investigation dashboard load hota hai.

💡 **Saral Bhasha Me:** Ye application ka gatekeeper hai. Investigators yaha se secure login karte hain aur demo testing ke liye single-click credentials provide kiye gaye hain.

3.2. Executive Dashboard ([#dashboard](#))

MISSION CONTROL

Purpose: Overall intelligence posture, prioritized leads, critical threat indicators aur live operational stats ka centralized overview.

- **Metric KPI Cards:** Total Targets Tracked, Active Operations (e.g. Operation Orion), High-Risk Alerts, Raw Signals Ingested.
- **Threat Posture Meter:** Real-time calculated threat severity level (Low, Moderate, Elevated, Critical).
- **Lead Prioritization Feed:** Highest risk suspect nodes jinpar immediate investigation action required hai.
- **Fast Actions:** One-click navigation to Network Analysis, Threat Map, or Report Generation.

💡 **Saral Bhasha Me:** Ye main control room hai jaha officer ko ek nazar me pata chalta hai ki kitne active threats hain aur kis suspect par pehle focus karna hai.

3.3. Investigation & Case Management ([#investigations](#))

CASE OPERATIONS

Purpose: Complex forensic investigations aur multi-agency operations ko track aur organize karna.

- **Case Board:** Har case ka Title (e.g. Operation Orion), Assigned Lead Investigator, Priority Level, Status (Open, In Progress, Under Review, Closed).
- **Case Actions:** "Create New Investigation", "Export Case Brief", "Link Additional Evidence / IOCs".
- **Cross-Entity Tagging:** Ek case me multiple suspect entities, compromised servers aur financial records ko link kiya ja sakta hai.

💡 **Saral Bhasha Me:** Yaha alag-alag criminal cases bante hain. Har case me kaun sa officer assigned hai aur case ki kya progress hai, sab track hota hai.

3.4. Global Threat Map & Geospatial Telemetry ([#map](#))

GEOSPATIAL INTELLIGENCE

Purpose: International cyber infrastructure, command & control (C2) servers, money laundering routes aur target locations ko world map par track karna.

- **Interactive Canvas Map:** High-performance canvas-based world map with dynamic zoom, panning, aur coordinate plotting.
- **Animated Trajectory Arcs:** Data exfiltration ya suspicious cross-border financial transactions ko pulsing arcs ke roop me dikhata hai.
- **Node Inspector Card:** Kisi bhi location marker par click karne par IP Address, ASN, Hosting Provider, Threat Score aur linked suspect info pop-up hoti hai.
- **Geospatial Filters:** Country-wise filter, Severity filter, Node-type filter (C2 Server, Proxy, Mule Account, Exit Node).

💡 **Saral Bhasha Me:** Ye live world map hai jisme dikhta hai ki cyber attacks ya funds kis country se kis country me travel kar rahe hain.

3.5. Entity Intelligence Registry & Deep Profile ([#entities](#) & [#entity](#))

TARGET REGISTRY

Purpose: Sabhi identified targets (individuals, hacker groups, crypto wallets, server infrastructure) ka master catalog.

- **Instant Search & Filter:** Target ID (e.g. ALPHA-17), Alias, Type ya Confidence level se real-time search.
- **Entity Profile Overview:**
 - **Threat Score Gauge:** Calculated risk rating (0 - 100%).
 - **IOC Matrix:** Linked IP addresses, Crypto addresses (BTC/ETH/USDT), Domains, Phone numbers.
 - **Known Aliases:** Resolved identities across different platforms (Dark web handles, Telegram IDs).
 - **Ingest IOC Action:** Naye indicators of compromise directly add karne ka workflow.

💡 **Saral Bhasha Me:** Ye suspects ki kundli / profile directory hai jisme unke sabhi fake names, IP addresses, phone aur crypto wallets ek jagah dikhte hain.

3.6. Interactive Network Graph & Link Analysis (#network)

GRAPH VISUALIZER

Purpose: Complex hidden relationships, communication loops aur financial syndicates ko visual force-directed graph me reveal karna.

- **Interactive Canvas / SVG Graph:** Real-time physics-based nodes and edges simulation (zoom, drag, node selection).
- **Multi-Hop Discovery:** 1-Hop, 2-Hop ya N-Hop relationship expansion to expose hidden intermediaries.
- **Bridge Node Detection:** Jo key operators alag-alag cyber criminal gangs ko connect karte hain unhe automatically flag karta hai.
- **Relationship Weight:** Edges par transaction volume, communication frequency ya confidence score display hota hai.

💡 **Saral Bhasha Me:** Ye visual web / connection graph hai. Kaun kiske sath baat kar raha hai ya paise bhej raha hai, ye visually connect karke dikhata hai.

3.7. Chronological Signal Timeline (#timeline)

TEMPORAL ANALYTICS

Purpose: Attack progression ya fraud sequence ko chronological order me step-by-step reconstruct karna.

- **Time-Sequenced Events:** Har signal ka exact timestamp, involved suspect, activity type aur severity rating.
- **Temporal Filtering:** Filter by date ranges, specific targets, ya event categories (e.g. Unauthorized Login, C2 Beacon, Wire Transfer).
- **Multi-Entity Overlay:** Ek sath do alag targets ke simultaneous actions ko compare karke coordination detect karna.

💡 **Saral Bhasha Me:** Ye incident ki time-machine hai jo batati hai ki pehle kya hua, fir hacker ne kya kiya aur kab fraud execute hua.

3.8. Multi-Source Intelligence Fusion (#fusion)

DATA INGESTION

Purpose: Raw unstructured aur structured data feeds ko ingest karke standardized intelligence me convert karna.

- **Supported Ingestion:** CSV/JSON logs, PCAP network metadata, threat feeds, manual evidence submissions.
- **Automated Pipeline:** Ingestion -> Data Cleaning -> Deduplication -> Entity Matching -> Graph Association.
- **Real-Time Stream Feed:** Live stream of ingested intelligence signals with validation flags.

💡 **Saral Bhasha Me:** Yaha alag-alag files aur logs upload hote hain jinhe system automatically analyze karke connect kar deta hai.

3.9. Threat Trends & Anomaly Analytics (#trends)

STATISTICAL ENGINE

Purpose: Large-scale trends, attack spikes aur anomalous patterns ko identify karna.

- **Trend Visualizations:** Daily signal volumes, high-severity anomaly bursts, target vulnerability distributions.
- **Pattern Recognition:** Coordinated botnet activity ya abnormal money transfer spikes ko auto-detect karta hai.

💡 **Saral Bhasha Me:** Ye graphical graphs aur charts dikhata hai ki threats kab badh rahe hain aur kaun sa pattern abnormal hai.

3.10. Real-Time Threat Alert Center (#alerts)

ACTIVE TRIAGING

Purpose: Machine learning aur rule-based triggers se generate hone wale critical alerts ka triage console.

- **Severity Levels:** Critical (Red), High (Orange), Medium (Yellow), Low (Blue).
- **Triage Workflow:** Acknowledge Alert, Escalate to New Case, Link to Existing Investigation, Dismiss False Positive.

💡 **Saral Bhasha Me:** Ye emergency warning center hai. Jaise hi koi high-risk event hota hai, yaha turant notification aati hai.

3.11. Evidence Locker & Chain of Custody (#evidence)

FORENSIC INTEGRITY

Purpose: Digital evidence items ko tamper-proof record aur cryptographic SHA-256 integrity ke sath manage karna.

- **Evidence Vault:** PCAP captures, server logs, financial ledgers, email headers with secure cryptographic checksums.
- **Chain of Custody Tracking:** Kis investigator ne kab evidence upload kiya, view kiya ya export kiya ka permanent log.

💡 **Saral Bhasha Me:** Court me pesh karne layak digital saboot ka locker jaha saboot ka hash aur timestamp safe rehta hai.

3.12. Structured Intelligence Reporting (#reports)

DOSSIER BUILDER

Purpose: Validated findings ko standard law enforcement aur executive intelligence dossiers me compile karna.

- **Dossier Generator:** Executive Summary, Suspect Matrix, Timeline Breakdown, Evidence Proof, aur Recommended Actions automatically compile karta hai.
- **Exporting:** Print-ready PDF report generation aur structured JSON export.

💡 **Saral Bhasha Me:** Pure case ki formal report aur PDF export karne ka module.

3.13. Audit Trails (#audit) & 3.14. TRACE-X AI Copilot Analyst

COMPLIANCE & DUAL AI

Audit Trails: Har user action, query, edit aur login ka tamper-evident log maintain karta hai legal compliance ke liye.

TRACE-X AI Copilot (Shortcut:  J ya Floating Icon):

- **Executive Briefing:** "Summarize Operation Orion" prompt se instant case summary deta hai.
- **Threat Scoring:** Highest risk targets aur bridge nodes identify karta hai.
- **Natural Language Query:** Database ke kisi bhi target ya IOC ke bare me conversational analysis provide karta hai.

💡 **Saral Bhasha Me:** Audit logs har activity ka hisaab rakhte hain, aur AI Copilot ek smart personal assistant hai jo case solve karne me madad karta hai.

4. Backend REST API Architecture & Endpoints

Backend Express.js server clean REST API architecture follow karta hai. Niche sabhi main endpoints ki details hain:

Endpoint Route	HTTP Method	Controller / Responsibility
/api/auth/login	POST	Investigator credentials verify karke JWT access token return karta hai.
/api/auth/signup	POST	Naya investigator account register karta hai with hashed password.
/api/entities	GET / POST	Tracked entities ki list fetch karta hai ya naya suspect/IOC add karta hai.
/api/entities/:id	GET / PUT	Specific target ka deep profile data, aliases aur threat score return karta hai.
/api/relationships	GET / POST	Network link graph ke liye source-to-target connections & edge types provide karta hai.
/api/signals	GET / POST	Multi-source telemetry signals fetch karta hai ya naye logs ingest karta hai.
/api/investigations	GET / POST	Active cases aur operations create aur track karta hai.
/api/evidence	GET / POST	Cryptographic hashes ke sath forensic artifacts aur chain-of-custody manage karta hai.
/api/alerts	GET / PUT	Real-time threat notifications fetch karta hai aur triage status update karta hai.
/api/ai/query	POST	AI Copilot queries process karke dynamic intelligence responses formulate karta hai.
/api/reports	GET / POST	Formal investigation reports generate aur retrieve karta hai.
/api/audit	GET	System compliance aur user activity audit logs provide karta hai.

5. Step-by-Step Local Setup & Execution

 **Quickest Method:** Use `start-tracex.bat`

Project directory me bani `start-tracex.bat` script par double-click karein. Ye automatically Backend aur Frontend dono ko parallel terminal me start kar degi!

Manual Terminal Instructions:

1. Backend Setup

```
cd Backend
npm install
npm run dev
```

Server `http://localhost:5000` par chalega with REST endpoints ready.

2. Frontend Setup

```
cd frontend
npm install
npm run dev
```

Vite dev server `http://localhost:5173` par launch ho jayega.

6. Key Viva & Presentation Talking Points

Q: TRACE-X traditional security tools se alag kaise hai?

Traditional tools isolated logs dikhate hain. TRACE-X multi-source signals ko fuse karke entity resolution, graph link analysis aur automated lead prioritization provide karta hai.

Q: Force-Directed Graph ka kya fayda hai?

Complex fraud syndicates me intermediary bridge operators hote hain jo directly connect nahi dikhte. Multi-hop graph analysis unhe instantly expose kar deta hai.

Q: Evidence Locker ki kya importance hai?

Forensic chain-of-custody aur SHA-256 cryptographic hashes ensure karte hain ki court of law me digital evidence tamper-proof aur legally valid rahe.

Q: AI Copilot kaise assist karta hai?

Investigator ko manually hazaro logs padhne ki zaroorat nahi padti. AI Copilot instant case briefings, threat rankings aur pattern anomalies summarize kar deta hai.